

Baseline values predict fasting glucose: a correlational analysis

Abstract

Background: A dietary intervention study on fasting glucose. Methods: The relationship between the baseline value and fasting glucose (mg/dL) at follow-up was quantified with the Pearson product-moment correlation coefficient, first in the full analysis sample and then within the mediterranean group only. Results: see the Results section. Conclusion: the findings are discussed in the context of the analysis sample.

Introduction

We report a dietary intervention study on fasting glucose. The primary question was whether the Mediterranean diet intervention is associated with fasting glucose (mg/dL) at follow-up. The dataset accompanying this report (data.csv) contains one row per participant and is described in the data dictionary.

Methods

Participants and data. Data were collected for 245 participants. Each record contains the variables listed in the data dictionary. Group membership is recorded in the column `diet`.

Exclusion criteria and preprocessing. Only participants aged 18 to 80 years inclusive were included in the analysis. Participants with missing values for glucose_fu were excluded (complete-case analysis). After the exclusions above, observations whose fasting glucose (mg/dL) at follow-up lay more than three sample standard deviations from the overall mean were treated as outliers and removed.

Statistical analysis. The relationship between the baseline value and fasting glucose (mg/dL) at follow-up was quantified with the Pearson product-moment correlation coefficient, first in the full analysis sample and then within the mediterranean group only. Two-sided p-values are reported. All analyses were performed on the analysis sample defined above. Statistics were computed with standard scientific software; no random resampling was used.

Results

Across all 216 participants, the baseline value correlated with the outcome ($r = 0.512$, $p < 0.001$). Within the mediterranean group ($n = 106$) the correlation was $r = 0.530$.

Discussion

The analysis followed the pre-specified plan described in the Methods. Limitations include the observational nature of some comparisons and the exclusion of records with missing values.